

Communication

A Simple Effective Permittivity Model for Metasurfaces within Multilayer Stratified Media

Filippo Costa, *Senior Member, IEEE*

Abstract—A simple and accurate approach for computing the effective permittivity of metasurfaces embedded within arbitrary stratified media is presented. The proposed model provides accurate estimation of the effective permittivity for stackups with several dielectric layers with arbitrary thickness and permittivity. The effect due to the unit cell geometry is also taken into account. The accuracy of the model is rigorously verified by comparing the effective permittivity estimated by the approximate model with the one extracted by using full-wave MoM simulations as a function of the substrate thickness.

Index Terms—Effective Permittivity, Equivalent Circuit Model, Frequency Selective Surface (FSS), Metasurfaces, Transmission Line (TL) Model.

I. INTRODUCTION

A powerful technique for analyzing the frequency response of Frequency Selective Surfaces (FSS) [1] or metasurfaces [2]–[4] is the equivalent circuit Transmission Line (TL) model [5] as it provides a useful starting point and a deep physical insight in the study of periodic surfaces [6]–[15]. In the transmission line model, the metasurface is represented by a lumped impedance [12], [16]–[18] and the L and C values represent a specific element shape. Analytic expressions for L and C elements are available for simple elements such as dense grids or array of patches [19]. Closed form relations for resonant elements (e.g. loops, crosses, Jerusalem crosses) are available but based on empirical parameters and they are often not sufficiently accurate. A more general approach consists in deriving the freestanding lumped parameters L_0 and C_0 by processing the metasurface impedance extracted after a freestanding full-wave simulation [5]. It is important to underline that the LC parameters representing the periodic surface are affected by the specific stratified dielectric configuration (number of dielectrics, thickness and dielectric permittivity). Indeed, the values of LC components need to be adjusted by multiplying capacitances and inductances by the effective dielectric permittivity or the effective magnetic permeability, respectively. For synthesis purposes, it is vital to derive a simple model for the effective permittivity of stratified dielectrics around the metasurface since this allows the derivation of the metasurface lumped parameters starting from their freestanding values. If the surrounding dielectrics are homogeneous and the dielectric thickness is large compared to the periodicity, the electric field is entirely contained within the

same dielectric and the effective permittivity is simply equal to the permittivity of the hosting medium [18]. However, when the thickness of the surrounding dielectrics is small compared to the lattice periodicity of the metasurface, the effective permittivity value is lower than the actual dielectric permittivity of the surrounding substrate [5], [20]. The presence of thin dielectric substrates close to the periodic surface involves a relevant number of harmonics [21]–[23] and a closed form relation for the effective permittivity is not available. In [5] a simple effective permittivity fitting model was provided. However, the formulation is accurate only for the case of two dielectrics surrounding a periodic surface. The use of multilayer substrates around a metasurface provides additional degrees of freedom in shaping its frequency and angular response. Some examples of devices based on metasurfaces embedded within multilayer media can be found in mantle cloaking [24], [25], wide impedance matching networks [26], [27], wideband electromagnetic absorbers [28], [29]. A model for the effective permittivity of metasurfaces embedded within multilayer stackup is also important for the application of metasurface as matching layer for the field penetration inside body [30] and for the design of chipless RFID sensors [31], [32]. The purpose of this paper is to present an improved and generalized fitting model for computing the effective permittivity of metasurfaces embedded within generic stratified media. The proposed approach is necessary for a fast synthesis of metasurface-based devices based on inhomogeneous stackups and it is also useful for understanding the physical mechanism of the wave interaction. The paper is organized as follow: in Section II the role of the effective permittivity in metasurface response is explained. In Section III, the generalized approach for computing the effective permittivity is presented. Section IV is dedicated to the verification of the model through a rigorous comparison with numerical results. Conclusions are reported in Section V.

II. ROLE OF THE EFFECTIVE PERMITTIVITY

Let us consider an electromagnetic plane wave travelling along z -direction and impinging on a metasurface embedded within a multilayer dielectric medium as shown in Fig. 1(a). The equivalent transmission line model of the stratified structure is shown in Fig. 1(b). The reflection and transmission coefficients of the metasurface can be computed according to classical transmission line theory with ABCD matrix approach [5], [33], [34]. The ABCD matrix includes the impedance of the metasurface, namely Z_{FSS} , expressed in terms of LC networks. The equivalent capacitance of the metasurface C

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Filippo Costa is with University of Pisa, Dipartimento di Ingegneria dell'Informazione, Pisa, Italy. (corresponding author, e-mail: filippo.costa@unipi.it)

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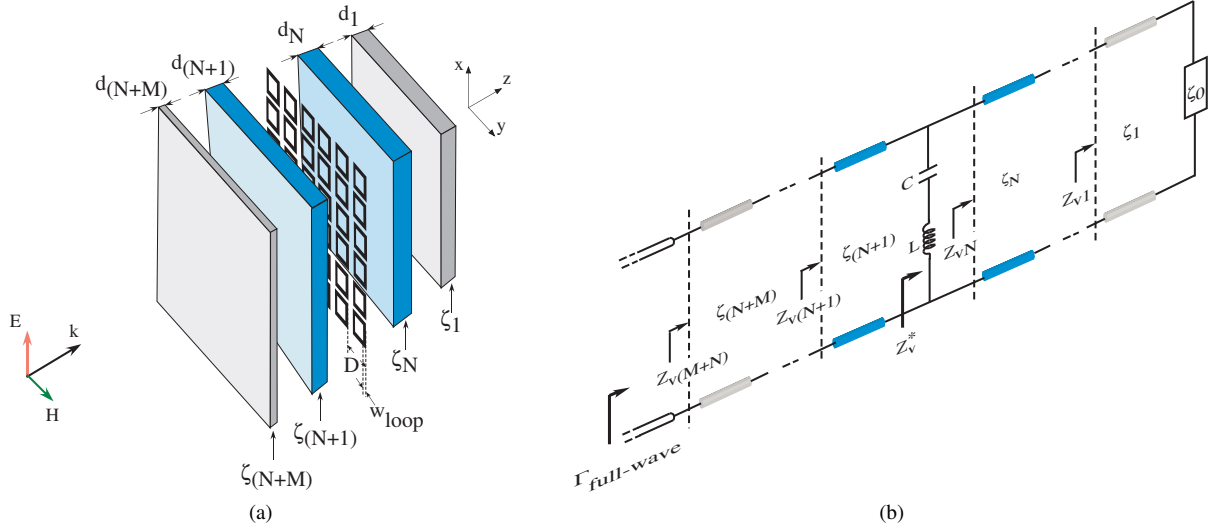


Fig. 1. (a) Layout of the FSS embedded within a generic multilayer. (b) Equivalent circuit of the electromagnetic problem.

should be modified proportionally to the effective dielectric permittivity due to surrounding media:

$$C = C_0 \varepsilon_{eff}(\varepsilon_{r1}, \dots, \varepsilon_{rn}, d_1, \dots, d_n, FSS_{geometry}) \quad (1)$$

where C_0 represents the unloaded capacitance computed in freestanding condition. A trivial approach for computing the effective permittivity consists in considering the average between the dielectric permittivities of surrounding substrates neglecting their thickness [18]. However, this approach is only reasonable for substrates thicker than approximately $D/3$, where D is the periodicity of the metasurface. In general, the effective permittivity is a function of several parameters: number of employed dielectrics, their dielectric permittivity and thickness and geometry of the periodic element.

A multilayer stackup affects the frequency response an FSS or a metasurface in a twofold way. The first effect is represented by the effective permittivity of the stackup which affects the FSS capacitance according to (1). The second effect is represented by the multiple reflections within the multilayer stackup which is taken into account by the transmission line model through the multiplication of the ABCD matrices of the dielectric layers.

The computation of the effective permittivity in case of multiple substrates is not trivial. In microstrip structures embedded within multiple dielectric layers a similar problem is considered and the effective permittivity is computed through the conformal mapping approach [35], [36] but for a specific geometry of the transmission line. Unfortunately, the aforementioned expressions, specifically computed for coplanar microstrip lines, do not adequately follow the effective permittivity variations in case of arbitrarily shaped metasurfaces.

An alternative approach consists in extracting the effective permittivity of a metasurface from numerical simulations and then deriving a suitable fitting model. According to (1), the effective permittivity can be computed as the ratio of the loaded capacitance, C , and the capacitance computed in the freestanding case, C_0 , as follow:

$$\varepsilon_{eff} = C/C_0 \quad (2)$$

The computation of the capacitance is performed by exploiting the transmission line model reported in Fig. 1(b). Once performed the freestanding simulation of the investigated structure, the reflection coefficient, $\Gamma_{full-wave}$, can be used to extract the input impedance as:

$$Z_{vM} = Z_0 \frac{1 + \Gamma_{full-wave}}{1 - \Gamma_{full-wave}} \quad (3)$$

where Z_0 represents the free-space impedance. By using classical transmission line theory [5], according to the equivalent transmission line model shown in Fig. 1b, the input impedance before the FSS layer can be derived as a function of the input impedance of the stackup Z_{vM} as:

$$Z_v^* = \frac{Z_{v(N+1)}\zeta_{(N+1)} - j\zeta_{(N+1)}^2 \tan(k_{z(N+1)}d_{N+1})}{\zeta_{(N+1)} - jZ_{v(N+1)} \tan(k_{z(N+1)}d_{N+1})} \quad (4)$$

where ζ_n, k_{zn} are the characteristic impedance, the normal component of the wavenumber and the thickness of the n^{th} layer. $Z_{v(n)}$ represents the input impedance before the n^{th} layer which can be derived by applying relation (4) iteratively. The impedance after the FSS layer can be derived, considering the simulated stackup, by using the transmission line theory iteratively. According to this approach, the impedance on the right hand side of the FSS layer can be expressed as:

$$Z_{vN} = \zeta_N \frac{Z_{v(N-1)} + j\zeta_N \tan(k_{zN}d_N)}{\zeta_N + jZ_{v(N-1)} \tan(k_{zN}d_N)} \quad (5)$$

Once derived Z_v^* and Z_{vN} , the FSS impedance can be easily computed as follows:

$$Z_{FSS} = \frac{(Z_v^* Z_{vN})}{(Z_{vN} - Z_v^*)} \quad (6)$$

In order to show the behaviour of the reflection coefficient as a function of the effective permittivity, an FSS comprising periodic square loops, embedded within two identical substrates

is simulated by using a Periodic Method of Moments (PMoM). As example, the dielectric permittivity of the two substrates is fixed to 5. The reflection coefficient of the FSS is reported in Fig. 2(a) for different values of the substrate thickness. A frequency shift is achieved by changing the thickness of the substrates. The surface impedance of the FSS as a function of the dielectric thickness of surrounding dielectrics is reported in Fig. 2(b). The full-wave results are compared with the results obtained by the transmission line approach based on the effective permittivity model. The thickness increase of the surrounding dielectrics determines a variation of the FSS resonance due to a change of the lumped capacitance according to (1). The physical reason of the change of effective permittivity is that the electric field around the periodic surface is not entirely contained within the substrate, thus the FSS capacitance decreases proportionally to the substrate thickness. The electric field distribution in the vicinity of the FSS is shown in Fig. 3 for different values of the substrates thickness. As is evident from Fig. 2(c), the thickness dependence can be neglected if the substrate is thicker than one third of the FSS periodicity.

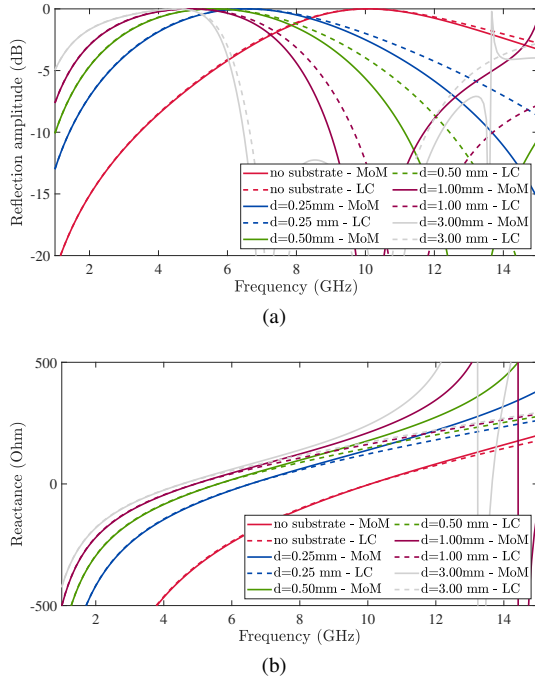


Fig. 2. (a) Reflection coefficient, (b) FSS surface impedance of a loop type FSS embedded within two identical dielectrics as a function of their thickness. $\epsilon_r = 5$. The loop FSS is characterized by a periodicity of $D = 10\text{mm}$ and a trace width (w_{loop}) of $D/16$.

III. GENERALIZED FITTING MODEL

The effective permittivity changes proportionally to the substrate thickness around the FSS and it approaches the dielectric permittivity of the surrounding substrate as its thickness reaches one third of the periodicity. A simple and accurate relation for fitting the transition function is [5]:

$$\epsilon_{eff} = \epsilon_r + (1 - \epsilon_r)e^{-\alpha(d/D)} \quad (7)$$

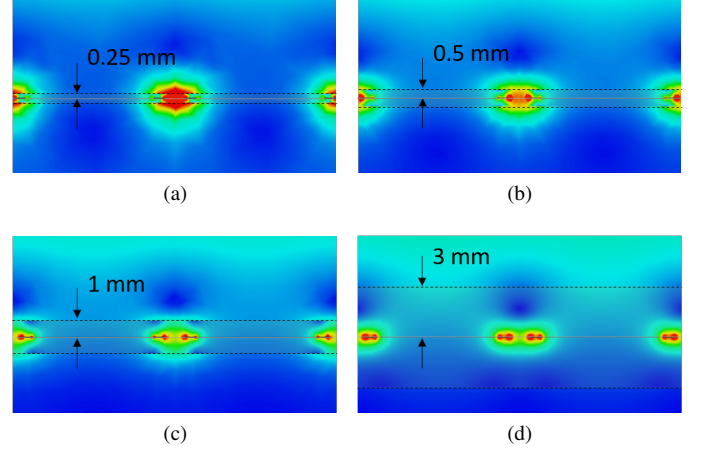


Fig. 3. Absolute value of the electric field distribution of a loop FSS with a periodicity of 10 mm embedded within two dielectric layers. The dielectric permittivity of the two layers is $\epsilon_r = 5$. (a) $d_1 = d_2 = 0.25\text{mm}$, (b) $d_1 = d_2 = 0.5\text{mm}$, (c) $d_1 = d_2 = 1\text{mm}$, (d) $d_1 = d_2 = 3\text{mm}$.

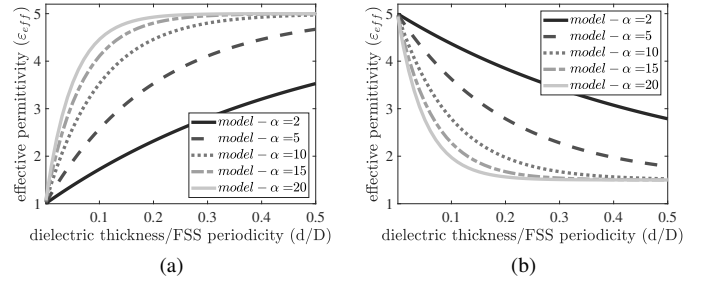


Fig. 4. Behaviour of the proposed transition function in (7) for different values of the shape factor α . (a) $\epsilon_r = 5$, $\epsilon_{r-ext} = 1$ (b) $\epsilon_r = 1$, $\epsilon_{r-ext} = 5$.

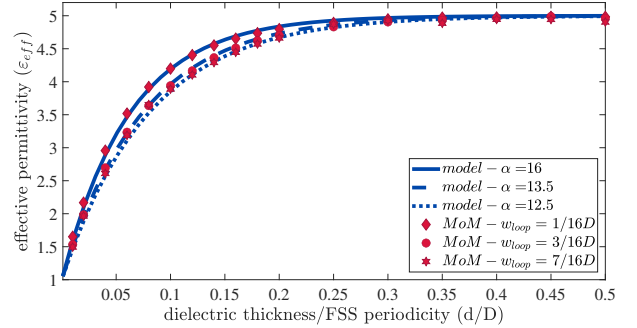


Fig. 5. Behaviour of the proposed effective permittivity for a loop type surface embedded within two dielectrics characterized by ϵ_r equal to 5. The effective permittivity of the loop computed for different width of the loop (varying of w) obtained by the MoM approach is compared with the behaviour proposed transition function by changing the value of the shape factor α .

where d is the thickness of the two substrates and D is the FSS periodicity. α is called *shape factor* since its value is a function of the geometry of the FSS unit cell. The relation (7) respects the following constraints given by the physics of the problem:

- $\epsilon_{eff} = 1$ if $d \rightarrow 0$
- $\epsilon_{eff} = \epsilon_r$ if $d/D \rightarrow \infty$
- $\epsilon_{eff} = 1$ if $\epsilon_r = 1$

Relation (7) is very accurate for the two substrates case

but it fails for more complex stackups. This work aims at generalizing the formulation.

A similar fitting relation can be also used if the external layer is denser than internal layer. In this case, the effective permittivity should decrease as the supporting dielectric thickness is enhanced. This is obtained by simply replacing the relative permittivity of the vacuum layer ($\varepsilon_r = 1$) with the relative permittivity of the most external dielectric ε_{r-ext} :

$$\varepsilon_{eff} = \varepsilon_r + (\varepsilon_{r-ext} - \varepsilon_r)e^{-\alpha(d/D)} \quad (8)$$

The behavior of the relations (7) and (8) is represented in Fig. 4 for two representative cases: (a) $\varepsilon_r = 1$ $\varepsilon_{r-ext} = 5$, (b) $\varepsilon_r = 5$ $\varepsilon_{r-ext} = 1$. The shape factor α is proportional to the filling factor of the unit cell [5]. As an example, a loop based periodic surface is analyzed as a function of the loop width, w_{loop} . In particular, in Fig. 5 the variation of the effective permittivity as a function of the substrate thickness is reported. In all cases the effective permittivity approaches the dielectric permittivity of the surrounding medium ($\varepsilon_r = 5$) but with a different slope of the curve. By selecting the appropriate value of the shape factor the numerical curves obtained by the MoM formulation can be opportunely fitted.

In order to generalize the evaluation of the effective permittivity for generic stackups, the global effective permittivity can be computed iteratively both for the right-hand side and for the left-hand side and finally averaged. For instance, in the case of N dielectrics on the right hand side of the FSS and M dielectrics on the left side of the FSS, the average effective permittivity is computed as:

$$\varepsilon_{eff} = \frac{\varepsilon_{eff-right}(N) + \varepsilon_{eff-left}(M)}{2} \quad (9)$$

where $\varepsilon_{eff-right}(N)$ and $\varepsilon_{eff-left}(M)$ terms are computed iteratively with the index n ranging from 1 to N and the index m ranging from 1 to M . Similarly to the relation (8), the effective permittivity of the n^{th} layer is written as a function of the $(n-1)^{th}$ layer as follow:

$$\varepsilon_{eff-right}(n) = \varepsilon_{rn} + (\varepsilon_{eff-right}(n-1) - \varepsilon_{rn})e^{-\alpha(d_n/D)} \quad (10)$$

$$\varepsilon_{eff-left}(m) = \varepsilon_{r(N+M-m+1)} + (\varepsilon_{eff-left}(m-1) - \varepsilon_{r(N+M-m+1)})e^{-\alpha(d_{(N+M-m+1)}/D)} \quad (11)$$

For the most external substrates the effective permittivity $\varepsilon_{eff-right}(n-1)$, $\varepsilon_{eff-left}(m-1)$ is the permittivity of the vacuum which is the most external dielectric:

$$\varepsilon_{eff-right}(1) = \varepsilon_{r1} + (1 - \varepsilon_{r1})e^{-\alpha(d_1/D)} \quad (12)$$

$$\varepsilon_{eff-left}(1) = \varepsilon_{rM} + (1 - \varepsilon_{rM})e^{-\alpha(d_M/D)} \quad (13)$$

The validity of the proposed approach is rigorously verified in the following section for a considerable number of representative and not trivial configurations. The verification

is performed by comparing the effective permittivity obtained according to relation (9) with the effective permittivity derived numerically by using MoM results.

IV. NUMERICAL RESULTS

The behaviour of the effective permittivity of a metasurface embedded within multilayered dielectric stackups is reported in this section for a set of representative examples. The results are obtained for a benchmark loop-shaped unit cell with the following geometrical parameters: $D = 10mm$ and $w_{loop} = 1/16D$. The validity of the proposed model is not anyway restricted to the specific unit cell geometry here analyzed since all the single resonant shapes can be taken into account by the previous formulation by adjusting the shape factor α appropriately. Initially, the case of a metasurface embedded within two dielectrics characterized by different geometrical and electrical properties is analyzed. Next, a metasurface embedded within 3 substrates is considered. For the three dielectrics stackup, two relevant cases are considered: inner layer characterized by a higher permittivity than the external one and the complementary configuration. The behaviour of the effective permittivity, according to the proposed generalized fitting model, is rigorously compared with the effective permittivity extracted by using the inversion procedure based on MoM simulations. For sake of completeness, the plots report also the behavior achieved with the previous formulation given in [5] for showing the relevant improvement of this approach.

A. Two dielectrics

The effective permittivity of the loop based metasurface as a function of the thickness of the two substrates characterized by different permittivity values ($\varepsilon_{r1} = 10.2$ and $\varepsilon_{r2} = 4.4$) is shown in Fig. 6(a). A simultaneous variation of the thickness of the two substrates (from $0.1mm$ to $10mm$) is considered. The effective permittivity grows from 1 (the vacuum dielectric permittivity) to the average value of the dielectric permittivity of the two media $(\varepsilon_{r1} + \varepsilon_{r2})/2 = 7.3$ as the thickness of the two layers becomes larger than $D/3$. The results of the fitting model agree very well with the effective permittivity computed with MoM simulations. Fig. 6(b) reports the two terms of the effective permittivity model used in relation (7). Both of the terms are characterized by a transition from the permittivity of the vacuum to the dielectric permittivity of the substrate. The grey curve represents the result obtained by using the relations in [5]. In this case, the two approaches are consistent. The results of the two approaches start diverging when the thicknesses of the substrates are different. To this aim, the effective permittivity of the same stackup is derived when only the thickness of one of the two substrates is varied. Fig. 7 reports the effective permittivity computed for a metasurface embedded within two different media as a function of the thickness of the layer 2 (the left side one). Two meaningful cases are reported in the plots: the first dielectric, characterized by the higher permittivity, is thick ($d_1 = 3mm$) or thin ($d_1 = 1mm$). As is evident from the plots, the proposed formulation provides accurate results differently from the previous fitting approach [5].

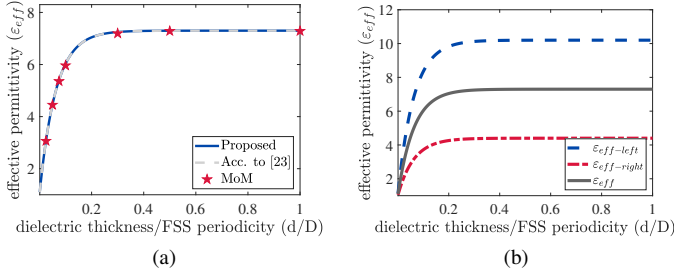


Fig. 6. Effective permittivity computed by using MoM approach and by using the proposed approximate relation. Two substrates with the following characteristics are used: $\epsilon_{r1} = 4.4$ $d_1/D = (0 : 1)$, $\epsilon_{r2} = 10.2$ and varying thickness. (a) global effective permittivity (ϵ_{eff}) (b) the two components of the average permittivity $\epsilon_{eff-right}$ and $\epsilon_{eff-left}$.

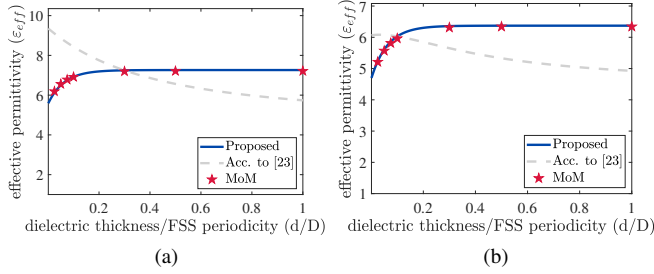


Fig. 7. Effective permittivity computed by using MoM approach and by using the approximate interpolating relation. Two substrates with the following characteristics are used: $\epsilon_{r1} = 4.4$ $d_1/D = (0 : 1)$, $\epsilon_{r2} = 10.2$ and (a) $d_2 = 3mm$, (b) $d_2 = 1mm$.

B. Three dielectrics

An additional interesting configuration is represented by an asymmetric dielectric stackup around the metasurface, with multiple dielectric substrates on one side of the periodic surface. In this configuration, two meaningful cases are possible: the external layer is denser than the internal one or the opposite scenario. Both cases have been analyzed by varying the thickness of the second dielectric of the stackup, that is, the one located between the periodic surface and the external dielectric on the right hand side. If the inner dielectric permittivity is lower than the external one, the increase of the thickness of layer 2 determines a decrease of the effective permittivity. This is due to a decrement of the term $\epsilon_{eff-right}$ while the thickness of layer 2 grows. The effective permittivity value gradually reaches the value of ϵ_{r2} . On the contrary, if the internal layer is characterized by a higher dielectric permittivity than layer 1, the term $\epsilon_{eff-right}$ increases gradually up to ϵ_{r2} . The behaviour of the effective permittivity for the case of a denser external medium, computed according to the proposed approach, is compared in Fig. 8(a) with the effective permittivity extracted from full-wave simulations. In Fig. 8(b) the terms $\epsilon_{eff-right}$ and $\epsilon_{eff-left}$ are reported together with their average value (ϵ_{eff}) in order to better clarify the physical behaviour of the effective permittivity. The effective permittivity in the case of a denser internal medium, computed according to the proposed approach, is compared in Fig. 9(a) with the effective permittivity extracted from full-wave simulations. In Fig. 9(b) the behaviour of terms $\epsilon_{eff-right}$ and $\epsilon_{eff-left}$ is

reported together with their average value (ϵ_{eff}). The most important difference between the two cases is the steepness of the transition which is much more rapid when the change is toward lower values. By looking at Fig. 8(a) and Fig. 9(a), it can be observed that the descent transition reaches its stability around $d/D = 0.15$ whereas the ascent transition reaches its stability around $d/D = 0.25$. This asymmetric effect can be taken into account in the fitting model by doubling the exponential number (2α for the descent transition). As is evident, the previous formulation [5] (dashed grey curve) does not correctly follow the behaviour of the effective permittivity for the analysed stackups.

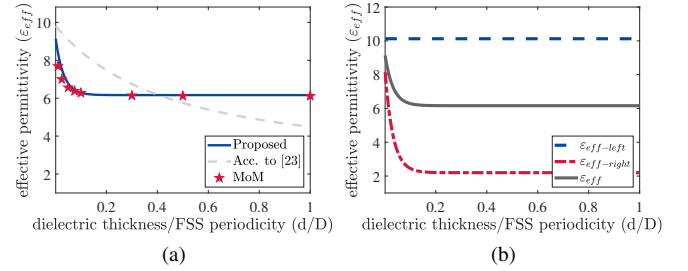


Fig. 8. Effective permittivity computed by using MoM simulations and by using the proposed approximate fitting model. The analyzed stackup comprises three substrates with the following characteristics: $\epsilon_{r1} = 10.2$ $d_1 = 1mm$, $\epsilon_{r2} = 2.2$ $d_2/D = (0 : 1)$, $\epsilon_{r3} = 10.2$, $d_3 = 3mm$. (a) effective permittivity ϵ_{eff} , (b) effective permittivity terms $\epsilon_{eff-right}$ and $\epsilon_{eff-left}$

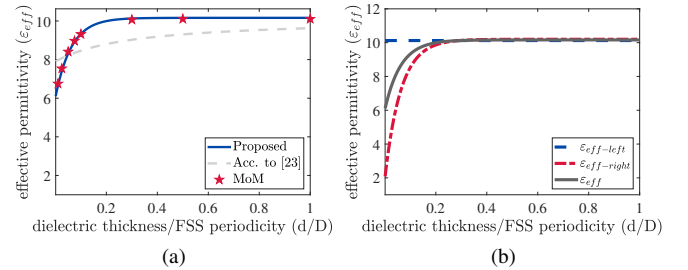


Fig. 9. Effective permittivity ϵ_{eff} computed by using MoM simulations and by using the proposed approximate fitting model. The analyzed stackup comprises three substrates with the following characteristics: $\epsilon_{r1} = 2.2$, $d_1 = 1mm$, $\epsilon_{r2} = 10.2$ $d_2/D = (0 : 1)$, $\epsilon_{r3} = 10.2$, $d_3 = 3mm$. (a) effective permittivity ϵ_{eff} , (b) effective permittivity terms $\epsilon_{eff-right}$ and $\epsilon_{eff-left}$

C. Additional dielectrics

Stacking additional dielectric layers complicates the analysed stackup and modifies the frequency response of the metasurface. However, the proposed formulation can also handle these more complex stackups according to relations (12) and (13). The adjustment of the capacitance through the proposed fitting model allows to reproduce the profile of the frequency reflection coefficient for arbitrary dielectric stackups. Therefore, after the initial characterization of a specific element in terms of LC parameters in freestanding configuration (L_0 and C_0), metasurfaces embedded within a generic dielectric stackup can be synthesized with a negligible computational effort. The computation time for the proposed model is a fraction of second ($t < 0.1s$ with 100 frequency points) and it is much faster than a MoM simulation that

requires a computation time proportional to the number of basic functions allocated for the discretization of the surface current ($t = 60s - 300s$ with 100 frequency points).

V. CONCLUSION

A simple but general approach for computing the effective permittivity of a metasurface embedded in arbitrary multilayer stratified media is presented. The validity of the presented formulation is rigorously verified by comparing the effective permittivity behaviour as a function of the substrate thickness obtained with the proposed model and the one numerically extracted after a MoM simulation. Some relevant examples comprising two or three layers stackups are analysed to show the accuracy of the proposed approach. The proposed fitting methodology allows the synthesis of metasurfaces, characterized in terms of LC parameters in a freestanding configuration, when embedded within arbitrary dielectric stackups.

REFERENCES

- [1] B. A. Munk, *Frequency selective surfaces: theory and design*. John Wiley & Sons, 2005.
- [2] S. B. Glybovski, S. A. Tretyakov, P. A. Belov, Y. S. Kivshar, and C. R. Simovski, "Metasurfaces: From microwaves to visible," *Physics reports*, vol. 634, pp. 1–72, 2016.
- [3] C. L. Holloway, E. F. Kuester, J. A. Gordon, J. O'Hara, J. Booth, and D. R. Smith, "An overview of the theory and applications of metasurfaces: The two-dimensional equivalents of metamaterials," *IEEE Antennas and Propagation Magazine*, vol. 54, no. 2, pp. 10–35, 2012.
- [4] M. Albooyeh, D.-H. Kwon, F. Capolino, and S. Tretyakov, "Equivalent realizations of reciprocal metasurfaces: Role of tangential and normal polarization," *Physical Review B*, vol. 95, no. 11, p. 115435, 2017.
- [5] F. Costa, A. Monorchio, and G. Manara, "Efficient analysis of frequency-selective surfaces by a simple equivalent-circuit model," *IEEE Antennas and Propagation Magazine*, vol. 54, no. 4, pp. 35–48, 2012.
- [6] C. S. Kaipa, A. B. Yakovlev, F. Medina, F. Mesa, C. A. Butler, and A. P. Hibbins, "Circuit modeling of the transmissivity of stacked two-dimensional metallic meshes," *Optics express*, vol. 18, no. 13, pp. 13 309–13 320, 2010.
- [7] E. Doumanis, G. Goussetis, J. L. Gomez-Tornero, R. Cahill, and V. Fusco, "Anisotropic impedance surfaces for linear to circular polarization conversion," *IEEE Transactions on Antennas and Propagation*, vol. 60, no. 1, pp. 212–219, 2011.
- [8] N. Behdad and M. A. Al-Joumayly, "A generalized synthesis procedure for low-profile, frequency selective surfaces with odd-order bandpass responses," *IEEE Transactions on Antennas and Propagation*, vol. 58, no. 7, pp. 2460–2464, 2010.
- [9] F. Costa and A. Monorchio, "Multiband electromagnetic wave absorber based on reactive impedance ground planes," *IET microwaves, antennas & propagation*, vol. 4, no. 11, pp. 1720–1727, 2010.
- [10] S. Ghosh and K. V. Srivastava, "An equivalent circuit model of fss-based metamaterial absorber using coupled line theory," *IEEE Antennas and Wireless Propagation Letters*, vol. 14, pp. 511–514, 2014.
- [11] Y. Zhao and A. Alù, "Manipulating light polarization with ultrathin plasmonic metasurfaces," *Physical Review B*, vol. 84, no. 20, p. 205428, 2011.
- [12] M. Selvanayagam and G. V. Eleftheriades, "Circuit modeling of huygens surfaces," *IEEE Antennas and Wireless Propagation Letters*, vol. 12, pp. 1642–1645, 2013.
- [13] G. Lovat, P. Burghignoli, and D. R. Jackson, "Fundamental properties and optimization of broadside radiation from uniform leaky-wave antennas," *IEEE Transactions on Antennas and Propagation*, vol. 54, no. 5, pp. 1442–1452, 2006.
- [14] A. Díaz-Rubio and S. A. Tretyakov, "Acoustic metasurfaces for scattering-free anomalous reflection and refraction," *Physical Review B*, vol. 96, no. 12, p. 125409, 2017.
- [15] F. Costa and A. Monorchio, "Closed-form analysis of reflection losses in microstrip reflectarray antennas," *IEEE transactions on antennas and propagation*, vol. 60, no. 10, pp. 4650–4660, 2012.
- [16] C. Pfeiffer and A. Grbic, "Bianisotropic metasurfaces for optimal polarization control: Analysis and synthesis," *Physical Review Applied*, vol. 2, no. 4, p. 044011, 2014.
- [17] S. Maci, M. Caiazzo, A. Cucini, and M. Casaletti, "A pole-zero matching method for ebg surfaces composed of a dipole fss printed on a grounded dielectric slab," *IEEE Transactions on Antennas and Propagation*, vol. 53, no. 1, pp. 70–81, 2005.
- [18] O. Luukkonen, C. Simovski, G. Granet, G. Goussetis, D. Lioubtchenko, A. V. Raisanen, and S. A. Tretyakov, "Simple and accurate analytical model of planar grids and high-impedance surfaces comprising metal strips or patches," *IEEE Transactions on Antennas and Propagation*, vol. 56, no. 6, pp. 1624–1632, 2008.
- [19] S. Tretyakov, *Analytical modeling in applied electromagnetics*. Artech House, 2003.
- [20] P. Callaghan, E. A. Parker, and R. J. Langley, "Influence of supporting dielectric layers on the transmission properties of frequency selective surfaces," in *IEE Proceedings H (Microwaves, Antennas and Propagation)*, vol. 138, no. 5. IET, 1991, pp. 448–454.
- [21] R. Dubrovka, J. Vazquez, C. Parini, and D. Moore, "Multi-frequency and multi-layer frequency selective surface analysis using modal decomposition equivalent circuit method," *IET microwaves, antennas & propagation*, vol. 3, no. 3, pp. 492–500, 2009.
- [22] M. Garcia-Vigueras, F. Mesa, F. Medina, R. Rodriguez-Berral, and J. L. Gomez-Tornero, "Simplified circuit model for arrays of metallic dipoles sandwiched between dielectric slabs under arbitrary incidence," *IEEE transactions on antennas and propagation*, vol. 60, no. 10, pp. 4637–4649, 2012.
- [23] S. Monni, G. Gerini, A. Neto, and A. G. Tijhuis, "Multimode equivalent networks for the design and analysis of frequency selective surfaces," *IEEE transactions on antennas and propagation*, vol. 55, no. 10, pp. 2824–2835, 2007.
- [24] E. G. Mizuji, A. Abdolali, and F. Aghamohamadi, "Analytical approach for making multilayer and inhomogeneous structures invisible based on mantle cloak," *Journal of Microwaves, Optoelectronics and Electromagnetic Applications*, vol. 17, no. 1, pp. 121–133, 2018.
- [25] A. Monti, J. C. Soric, A. Alu, A. Toscano, and F. Bilotti, "Anisotropic mantle cloaks for tm and te scattering reduction," *IEEE Transactions on Antennas and Propagation*, vol. 63, no. 4, pp. 1775–1788, 2015.
- [26] E. Magill and H. Wheeler, "Wide-angle impedance matching of a planar array antenna by a dielectric sheet," *IEEE Transactions on Antennas and Propagation*, vol. 14, no. 1, pp. 49–53, 1966.
- [27] D. Bianchi, S. Genovesi, M. Borgese, F. Costa, and A. Monorchio, "Element-independent design of wide-angle impedance matching radomes by using the generalized scattering matrix approach," *IEEE Transactions on Antennas and Propagation*, vol. 66, no. 9, pp. 4708–4718, 2018.
- [28] M. Cao, B. Wang, Q. Li, J. Yuan, G. Xu, S. Qin, and X. Fang, "Towards an intelligent cad system for multilayer electromagnetic absorber design," *Materials & design*, vol. 19, no. 3, pp. 113–120, 1998.
- [29] S. Chakravarty, R. Mittra, and N. R. Williams, "On the application of the microgenetic algorithm to the design of broad-band microwave absorbers comprising frequency-selective surfaces embedded in multilayered dielectric media," *IEEE Transactions on Microwave Theory and Techniques*, vol. 49, no. 6, pp. 1050–1059, 2001.
- [30] S. Genovesi, I. R. Butterworth, J. E. C. Serrallés, and L. Daniel, "Metasurface matching layers for enhanced electric field penetration into the human body," *IEEE Access*, vol. 8, pp. 197 745–197 756, 2020.
- [31] F. Costa, A. Gentile, S. Genovesi, L. Buoncristiani, A. Lazaro, R. Villarino, and D. Girbau, "A depolarizing chipless rf label for dielectric permittivity sensing," *IEEE Microwave and Wireless Components Letters*, vol. 28, no. 5, pp. 371–373, 2018.
- [32] N. C. Karmakar, E. M. Amin, and J. K. Saha, *Chipless RFID sensors*. John Wiley & Sons, 2016.
- [33] T. Zhao, D. R. Jackson, J. T. Williams, and A. Oliner, "Simple cad model for a dielectric leaky-wave antenna," *IEEE Antennas and Wireless Propagation Letters*, vol. 3, no. 1, pp. 243–245, 2004.
- [34] M. Borgese and F. Costa, "A simple equivalent circuit approach for anisotropic frequency selective surfaces," *IEEE Transactions on Antennas and Propagation*, vol. –, no. –, pp. –, accepted for publication.
- [35] S. S. Gevorgian, "Basic characteristics of two layered substrate coplanar waveguides," *Electronics Letters*, vol. 30, no. 15, pp. 1236–1237, 1994.
- [36] E. Carlsson and S. Gevorgian, "Conformal mapping of the field and charge distributions in multilayered substrate cpws," *IEEE Transactions on Microwave Theory and Techniques*, vol. 47, no. 8, pp. 1544–1552, 1999.